Shanmugam Subramanian

Mechanical Engineer II | Mechatronics & Electromechanical Design | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience designing electromechanical products that combine RF PCBs, sheet metal, molded and cast parts for 5G antenna and automotive radar hardware. Developed a pneumatic punching fixture (yield 60% to 95%) and built a 6-DOF robotic arm and 1:64 gearbox as personal projects. Experienced in FEA, and fixture and automation design.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

Electromechanical Design, Fixtures & Automation

- Developed a **pneumatic pivot punching fixture** cutting a 2-person task to one operator & **raising yield from 60% to 95%**.
- Built RF PCB fixtures, press-pull tools, & alignment jigs: **40% less rework, 75% faster riveting**, 20–50 min saved per unit.
- Designed pole, wall, ceiling, & tri-sector mounting brackets; gravity-assisted design **halved installation manpower**.
- Built Excel-based **hand-calculation tools** for drag force, pneumatic crimping force, & plastic snap-joint behavior.
- Introduced topology-optimized FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.
- Authored a **utility patent draft** for a snap-fit PCB/sheet-metal support pillar (twist-lock base, two-cavity mold).
- Drove **simulation-led design** (snap-fit, topology optimization, structural FEA) for 3 injection-molded parts & 1 pultruded FRP radome: **first-pass success** on all, ∼**$15K under tooling budget**.

Product Development & NPI

- Led **electro-mechanical packaging** of eight 5G antenna products: released RF PCB, sheet metal, die-cast, CNC, & EDM parts with a >**90% FAI pass rate**.
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.

CAE & Validation

- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Performed **IEC 60068-2-31 drop tests** for mechanical qualification of antenna products.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.
- Created company-wide central library in SolidWorks with 700+ frequently used fasteners, reducing design lead time.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

6-DOF Mimicking Robot— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.
Gearbox design — Dec 2025 Personal project
Designed and built a 1:64 multi-stage spur gear gearbox system to power a conveyor platform for scaled vehicle testing.
Hole Anchor Gripper Personal project
Designed a cam-lever expanding anchor gripper that inserts into holes and locks to lift loads.
SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing, mechanism & gear design, pneumatic fixture design
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis
Programming & Embedded systems: Mechatronics: Python, MATLAB Simulink & Simscape, Arduino, C, C++
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Management & Quality: RCA, Risk Mitigation, Feasibility Analysis, Cross-functional Team Coordination